



Optimal Upper Estimates for the First Eigenvalue of a Jacobi Type Operator in Spherical and Hyperbolic Spaces

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Abstract. In this paper, our aim is to establish optimal upper estimates for the first positive eigenvalue of a Jacobi type operator, which is a suitable extension of the linearized operators of the higher order mean curvatures of a closed hypersurface immersed either in spherical or in hyperbolic spaces.

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1. Introduction

Into the scope of geometric analysis, an important theme is the study of the first positive eigenvalue of certain elliptic operators defined on Riemannian manifolds and we can truly say that it was initiated by Reilly in the seminal paper [11], where he established some sharp upper estimates concerning the first positive eigenvalue of the Laplacian operator of a closed hypersurface immersed in the Euclidean space.

Continuing in this branch, many authors presented generalizations and extensions of Reilly's inequalities to some other ambient spaces (we refer, for instance, the articles [4, 7, 9, 10, 13]). Among such works, we point out that, simultaneously and independently, Alencar, do Carmo and Marques [2] and Grosjean [9] obtained optimal upper bounds for the first positive eigenvalue $\lambda_1^{L_r}$ of the linearized operator L_r of the r th mean curvature H_r of a closed hypersurface immersed either in spherical or in hyperbolic spaces.

Our aim in this paper is to study the behavior of the first positive eigenvalue $\lambda_1^{\mathcal{L}_{r,s}}$ of the Jacobi type (or simply, Jacobi) operator $\mathcal{L}_{r,s}$ which is defined as follows: fixed integer numbers r, s such that $0 \leq r \leq s \leq n-1$, $\mathcal{L}_{r,s} : C^\infty(M) \rightarrow C^\infty(M)$ is given by

$$\mathcal{L}_{r,s}(f) = \sum_{j=r}^s (j+1)a_j L_j(f), \quad (1.1)$$

where L_j are the linearized operators of the j th mean curvatures H_j , a_j are nonnegative real numbers (with at least one nonzero) for all $j \in \{r, \dots, s\}$ and f is a smooth function defined on a closed hypersurface M^n which is supposed immersed either in spherical or in hyperbolic spaces. In this setting, we prove optimal upper estimates for $\lambda_1^{\mathcal{L}_{r,s}}$ extending previous ones obtained in [2, 9] (see Theorems 1–3).

We also observe that the authors in [8] established the notion of (r, s) -stability concerning closed hypersurfaces with higher order mean curvatures linearly related in a Riemannian space form. In particular, they obtained a suitable characterization of the (r, s) -stability through the analysis of the first positive eigenvalue $\lambda_1^{\mathcal{L}_{r,s}}$ of the Jacobi operator $\mathcal{L}_{r,s}$, which is associated with the corresponding variational problem (cf. Theorem 5.3 of [8]). Our purpose in this work is exactly to obtain optimal upper estimates for $\lambda_1^{\mathcal{L}_{r,s}}$. Consequently, the results that we will present along this paper are naturally related to the study of (r, s) -stable closed hypersurfaces in a Riemannian space form.

2. Preliminaries

Given a connected and orientable hypersurface $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ into a Riemannian space form of constant sectional curvature δ , one can choose a globally defined unit normal vector field N on M^n . Let A denote the shape operator with respect to N , so that, at each $p \in M^n$, A restricts to a self-adjoint linear map $A_p : T_p M \rightarrow T_p M$.

For $1 \leq r \leq n$, if we let $S_r(p)$ denote the r th elementary symmetric function on the eigenvalues of A_p , we get n smooth functions $S_r : M^n \rightarrow \mathbb{R}$ such that

$$\det(tI - A) = \sum_{k=0}^n (-1)^k S_k t^{n-k},$$

where $S_0 = 1$ by construction. If $p \in M^n$ and $\{e_1, \dots, e_n\}$ is an orthonormal basis of $T_p M$ formed by eigenvectors of A_p , with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$, one immediately sees that

$$S_r = \sigma_r(\lambda_1, \dots, \lambda_n),$$

where $\sigma_r \in \mathbb{R}[X_1, \dots, X_n]$ is the r th elementary symmetric polynomial on the indeterminates $X_1, \dots, X_n$.

For $1 \leq r \leq n$, one defines the r th mean curvature H_r of $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ by

$$\binom{n}{r} H_r = S_r = S_r(\lambda_1, \dots, \lambda_n).$$

In particular, for $r = 1$,

$$H_1 = \frac{1}{n} \sum_{k=1}^n \lambda_k$$

is exactly the *mean curvature* H of x .

We also define, for $0 \leq r \leq n$, the r th *Newton transformation* P_r on M^n by setting $P_0 = I$ (the identity operator) and, for $1 \leq r \leq n$, via the recurrence relation

$$P_r = S_r I - A P_{r-1}.$$

From Lemma 2.1 of [5], we have the following trace formulas:

$$\begin{cases} \operatorname{tr}(P_r) = b_r H_r; \\ \operatorname{tr}(A P_r) = b_r H_{r+1}; \\ \operatorname{tr}(A^2 P_r) = n \frac{b_r}{r+1} H_1 H_{r+1} - b_{r+1} H_{r+2}, \end{cases} \quad (2.1)$$

where $b_r = (r+1) \binom{n}{r+1} = (n-r) \binom{n}{r}$.

For what follows, let s_δ be the solution of the Cauchy's problem

$$\begin{cases} y''(t) + \delta y(t) = 0, \\ y(0) = 0, y'(0) = 1. \end{cases} \quad (2.2)$$

If we establish the notation $c_\delta = s'_\delta$, then $c'_\delta = -\delta s_\delta$ and $c_\delta^2 + \delta s_\delta^2 = 1$. We have $s_\delta(t) = \frac{1}{\sqrt{-\delta}} \sinh(\sqrt{-\delta}t)$ and $c_\delta(t) = \cosh(\sqrt{-\delta}t)$ if $\delta < 0$, and $s_\delta(t) = \frac{1}{\sqrt{\delta}} \sin(\sqrt{\delta}t)$ and $c_\delta(t) = \cos(\sqrt{\delta}t)$ if $\delta > 0$.

In this context, if $X = s_\delta \bar{\nabla} d$ is the position vector of M^n in $\bar{M}_\delta^{n+1}$ with respect to a point $p_0 \in \bar{M}_\delta^{n+1}$, where $d = d(p_0, \cdot)$ is the distance function from p_0 in $\bar{M}_\delta^{n+1}$, then

$$X^\top = s_\delta \nabla d \quad \text{and} \quad \nabla c_\delta = -\delta X^\top. \quad (2.3)$$

The following result corresponds to Lemma 2.2 of [2] and will be very useful to obtain our main results.

Lemma 1. *Let $x : M^n \rightarrow \bar{M}_\delta^{n+1}$ be a hypersurface immersed in a Riemannian space form $\bar{M}_\delta^{n+1}$ with constant sectional curvature δ . In the case $\delta > 0$, suppose that $x(M)$ is contained in convex ball of radius $\frac{\pi}{4\sqrt{\delta}}$ and center $p_0 \in \bar{M}_\delta^{n+1}$. Then,*

$$\sum_{i=1}^{n+1} \left\langle P_r \nabla \left(\frac{s_\delta(d)}{d} x_i \right), \nabla \left(\frac{s_\delta(d)}{d} x_i \right) \right\rangle + \delta \langle P_r(X^\top), X^\top \rangle = b_r H_r,$$

where s_δ is the solution of (2.2), $x_1, \dots, x_{n+1}$ are the normal coordinates centered in p_0 with respect to some orthonormal basis in $T_{p_0} \bar{M}$, $X^\top$ is vector field defined in (2.3) and $b_r = (r+1) \binom{n}{r+1}$.

Associated with each Newton transformation P_r , one has the *second-order linear differential operator* $L_r : C^\infty(M) \rightarrow C^\infty(M)$ given by

$$L_r(f) = \text{tr}(P_r \circ \text{Hess}f),$$

for every $f \in C^\infty(M)$.

We remark that L_0 is the Laplacian operator Δ , and L_1 is the Cheng-Yau's square operator $\square$ defined in [6]. According to [12], P_r is divergence-free whenever $\overline{M}_\delta^{n+1}$ is of constant sectional curvature; consequently,

$$L_r(f) = \text{div}(P_r \nabla f). \quad (2.4)$$

Useful consequences of (2.4) are given in the following result:

Lemma 2. *Let $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ be a closed hypersurface immersed in a Riemannian space form $\overline{M}_\delta^{n+1}$. Then, for all $f \in C^\infty(M)$ it holds*

$$\int_M L_r(f) dM = 0, \quad \int_M f L_r(f) dM = - \int_M \langle P_r \nabla f, \nabla f \rangle dM.$$

From Proposition 3.2 of [5], we have the following sufficient ellipticity criteria for the L_r operators.

Lemma 3. *Let $\overline{M}_\delta^{n+1}$ be either an open hemisphere of a spherical space (when $\delta > 0$) or a hyperbolic space (when $\delta < 0$), and let $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ be a closed hypersurface. If $H_{r+1} > 0$, then L_j is elliptic and $H_j > 0$, for all $j \in \{1, \dots, r\}$.*

3. Optimal Upper Estimates for $\lambda_1^{\mathcal{L},s}$

We begin this section by reasoning as in the proof of Lemma 2.3 of [2] to get

Proposition 1. *Let $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ be an orientable closed hypersurface immersed in a Riemannian space form $\overline{M}_\delta^{n+1}$ of constant sectional curvature δ . If $H_{s+1} > 0$, for some integer number $s \in \{1, \dots, n-1\}$, and $c_\delta \geq 0$, then*

$$\int_M c_\delta dM \leq \varepsilon_{r,s} \int_M s_\delta dM, \quad (3.1)$$

for all $r \in \{0, \dots, s\}$, where

$$\varepsilon_{r,s} = \frac{\sum_{j=r}^s (j+1) a_j b_j \left(\max_M H_{j+1} \right)}{\sum_{j=r}^s (j+1) a_j b_j \left(\min_M H_j \right)}, \quad b_j = (j+1) \binom{n}{j+1},$$

a_j are nonnegative real numbers (with at least one nonzero), for all $j \in \{r, \dots, s\}$.

Proof. Since $H_{s+1} > 0$, Lemma 3 guarantees us that $H_j > 0$ for all $j \in \{1, \dots, s\}$. Now, from Lemma 1 of [1], we have

$$L_j(c_\delta) = -\delta b_j (c_\delta H_j + \langle X, N \rangle H_{j+1}). \quad (3.2)$$

Taking into account that M^n is closed, from (3.2) and using the divergence theorem we obtain the following Minkowski integral formula for the $\mathcal{L}_{r,s}$ operator

$$\sum_{j=r}^s (j+1)a_j b_j \int_M (c_\delta H_j + \langle X, N \rangle H_{j+1}) dM = 0, \quad (3.3)$$

where $a_r, \dots, a_s$ are nonnegative real numbers with at least one nonzero.

On the other hand, using Cauchy–Schwarz inequality jointly with the fact that $X = s_\delta \bar{\nabla} d$, we have

$$-\langle X, N \rangle \leq |X||N| = |X| = s_\delta. \quad (3.4)$$

From (3.3) and (3.4), we obtain

$$\begin{aligned} \sum_{j=r}^s (j+1)a_j b_j \int_M c_\delta H_j dM &= - \sum_{j=r}^s (j+1)a_j b_j \int_M \langle X, N \rangle H_{j+1} dM \\ &\leq \sum_{j=r}^s (j+1)a_j b_j \int_M H_{j+1} |X| dM \\ &= \sum_{j=r}^s (j+1)a_j b_j \int_M H_{j+1} s_\delta dM \\ &\leq \sum_{j=r}^s (j+1)a_j b_j \left(\max_M H_{j+1} \right) \int_M s_\delta dM. \end{aligned} \quad (3.5)$$

Since $c_\delta \geq 0$, we have

$$\sum_{j=r}^s (j+1)a_j b_j \left(\min_M H_j \right) \int_M c_\delta dM \leq \sum_{j=r}^s (j+1)a_j b_j \int_M c_\delta H_j dM. \quad (3.6)$$

Hence from (3.5) and (3.6), we obtain the desired result. $\square$

Following the ideas of Alencar, do Carmo and Marques [2] and Grosjean [9], we can consider a convex ball of $\overline{M}_\delta^{n+1}$ to obtain a point $p_0 \in \overline{M}_\delta^{n+1}$ such that

$$\int_M \frac{s_\delta(d)}{d} x_i dM = 0, \quad (3.7)$$

for every $i \in \{1, \dots, n+1\}$, where $d = d(p_0, \cdot)$ is the distance function from p_0 in $\overline{M}_\delta^{n+1}$. In what follows, we will assume the existence of such a point.

Our first result is concerning closed hypersurfaces in a hyperbolical space, that is, a Riemannian space form with constant negative sectional curvature.

Theorem 1. *Let $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ be an orientable closed hypersurface immersed in a Riemannian space form $\overline{M}_\delta^{n+1}$ with constant sectional curvature $\delta < 0$. If $H_{s+1} > 0$, for some integer number $s \in \{1, \dots, n-1\}$, then*

$$\lambda_1^{\mathcal{L}_{r,s}} \leq (\delta + \varepsilon_{r,s}^2) \sum_{j=r}^s (j+1) a_j b_j \left(\min_M H_j \right), \quad (3.8)$$

for all $r \in \{0, \dots, s\}$, where

$$\varepsilon_{r,s} = \frac{\sum_{j=r}^s (j+1) a_j b_j \left(\max_M H_{j+1} \right)}{\sum_{j=r}^s (j+1) a_j b_j \left(\min_M H_j \right)}, \quad b_j = (j+1) \binom{n}{j+1}.$$

Here, $\lambda_1^{\mathcal{L}_{r,s}}$ is the first positive eigenvalue of the operator $\mathcal{L}_{r,s}$ defined in (1.1) and a_j are nonnegative real numbers (with at least one nonzero) for all $j \in \{r, \dots, s\}$. Moreover, the equality occurs in (3.8) if, and only if, $x(M)$ is a sphere of $\overline{M}_\delta^{n+1}$ and x is its inclusion as a geodesic sphere.

Proof. Since M^n is closed and $H_{s+1} > 0$, Lemma 3 guarantees that L_j is elliptic for $j \in \{1, \dots, s\}$ and, hence, $\mathcal{L}_{r,s}$ is elliptic. Thus, it holds the following characterization of $\lambda_1^{\mathcal{L}_{r,s}}$ through the Rayleigh quotient

$$\lambda_1^{\mathcal{L}_{r,s}} = \inf \left\{ \frac{-\int_M f \mathcal{L}_{r,s}(f) dM}{\int_M f^2 dM}; \int_M f dM = 0 \right\}. \quad (3.9)$$

For every $1 \leq i \leq n+1$, we denote $f_i = \frac{s_\delta(d)}{d} x_i$. From (3.7), we have that $\int_M f_i dM = 0$ and, using (3.9), we obtain

$$\lambda_1^{\mathcal{L}_{r,s}} \int_M f_i^2 dM \leq - \int_M f_i \mathcal{L}_{r,s}(f_i) dM. \quad (3.10)$$

Moreover, from Lemma 2 and (3.10), we have

$$\begin{aligned} \lambda_1^{\mathcal{L}_{r,s}} \int_M f_i^2 dM &\leq - \sum_{j=r}^s (j+1) a_j \int_M f_i L_j(f_i) dM \\ &= \sum_{j=r}^s (j+1) a_j \int_M \langle P_j(\nabla f_i), \nabla f_i \rangle dM. \end{aligned} \quad (3.11)$$

Now, summing on i from 1 to $n+1$ the expression (3.11) and using Lemma 1 we obtain

$$\sum_{i=1}^{n+1} f_i^2 = s_\delta^2(d) \quad \text{and} \quad \sum_{i=1}^{n+1} \langle P_j(\nabla f_i), \nabla f_i \rangle + \delta \langle P_j(X^\top), X^\top \rangle = b_j H_j. \quad (3.12)$$

Hence, joining (3.11) with (3.12), we get

$$\begin{aligned} \lambda_1^{\mathcal{L}_{r,s}} \int_M s_\delta^2 dM &\leq \sum_{j=r}^s (j+1) a_j \int_M \sum_{i=1}^{n+1} \langle P_j(\nabla f_i), \nabla f_i \rangle dM \\ &= \sum_{j=r}^s (j+1) a_j b_j \int_M H_j dM \\ &\quad - \delta \sum_{j=r}^s (j+1) a_j \int_M \langle P_j(X^\top), X^\top \rangle dM. \end{aligned} \quad (3.13)$$

On the one hand, from Lemma 2 and (2.3), we have

$$\int_M c_\delta L_j(c_\delta) dM = - \int_M \langle P_j(\nabla c_\delta), \nabla c_\delta \rangle dM = -\delta^2 \int_M \langle P_j(X^\top), X^\top \rangle dM. \quad (3.14)$$

From (3.13) and (3.14), we conclude that

$$\lambda_1^{\mathcal{L}_{r,s}} \int_M s_\delta^2 dM \leq \sum_{j=r}^s (j+1) a_j b_j \int_M H_j dM + \frac{1}{\delta} \sum_{j=r}^s (j+1) a_j \int_M c_\delta L_j(c_\delta) dM. \quad (3.15)$$

Hence from (3.2), (3.4) and (3.15), we obtain

$$\begin{aligned} \lambda_1^{\mathcal{L}_{r,s}} \int_M s_\delta^2 dM &\leq \sum_{j=r}^s (j+1) a_j b_j \int_M H_j dM \\ &\quad - \frac{1}{\delta} \sum_{j=r}^s (j+1) a_j \int_M c_\delta (\delta b_j H_j c_\delta + \delta b_j \langle X, N \rangle H_{j+1}) dM \\ &\leq \sum_{j=r}^s (j+1) a_j b_j \int_M (1 - c_\delta^2) H_j dM \\ &\quad + \sum_{j=r}^s (j+1) a_j b_j \left(\max_M H_{j+1} \right) \int_M s_\delta c_\delta dM \\ &= \delta \sum_{j=r}^s (j+1) a_j b_j \int_M s_\delta^2 H_j dM \\ &\quad + \sum_{j=r}^s (j+1) a_j b_j \left(\max_M H_{j+1} \right) \int_M s_\delta c_\delta dM. \end{aligned} \quad (3.16)$$

Now, if $\delta \leq 0$, Lemma 2.8 of [10], assures us that

$$\left(\int_M s_\delta dM \right) \left(\int_M s_\delta c_\delta dM \right) \leq \left(\int_M s_\delta^2 dM \right) \left(\int_M c_\delta dM \right). \quad (3.17)$$

Thus, from Proposition 1 and (3.17), we obtain

$$\int_M s_\delta c_\delta dM \leq \frac{\int_M c_\delta dM}{\int_M s_\delta dM} \int_M s_\delta^2 dM \leq \varepsilon_{r,s} \int_M s_\delta^2 dM. \quad (3.18)$$

Consequently, since $\delta < 0$, replacing (3.18) in (3.16), we have

$$\begin{aligned} \lambda_1^{\mathcal{L}_{r,s}} \int_M s_\delta^2 dM &\leq \delta \sum_{j=r}^s (j+1) a_j b_j \int_M s_\delta^2 H_j dM \\ &\quad + \varepsilon_{r,s}^2 \sum_{j=r}^s (j+1) a_j b_j \left(\min_M H_j \right) \int_M s_\delta^2 dM \\ &\leq (\delta + \varepsilon_{r,s}^2) \sum_{j=r}^s (j+1) a_j b_j \left(\min_M H_j \right) \int_M s_\delta^2 dM. \end{aligned} \quad (3.19)$$

Since $\int_M s_\delta^2 dM > 0$, from (3.19), we conclude that (3.8) holds.

Moreover, if equality occurs in (3.8), then necessarily also occurs the equality in (3.4), that is

$$-\langle X, N \rangle = |X| |N|,$$

with $X = s_\delta(d) \bar{\nabla} d$, what implies that $\bar{\nabla} d$ is orthogonal to $x(M)$. Thus, d is constant on $x(M)$ and, therefore, $x(M)$ is a sphere of $\mathbb{R}^{n+1}$ around p_0 and x is its inclusion as a geodesic sphere. $\square$

Now, let $\overline{M}_\delta^{n+1}$ be a spherical space, that is, a Riemannian space form with constant sectional curvature $\delta > 0$. According to [13], we will consider the *center of gravity* $\mathbf{c}$ of an orientable closed hypersurface $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ which is given by

$$\mathbf{c} = \frac{1}{\text{vol}(M)} \int_M c_\delta dM. \quad (3.20)$$

In this setting, we get

$$\int_M \frac{c_\delta - \mathbf{c}}{\sqrt{\delta}} dM = 0, \quad (3.21)$$

where $c_\delta = s'_\delta$. Moreover, from (3.20) we have that $|\mathbf{c}| < 1$.

Proposition 2. *Let $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ be an orientable closed hypersurface immersed in a Riemannian space form $\overline{M}_\delta^{n+1}$ with constant sectional curvature $\delta > 0$. If $H_{s+1} > 0$, for some integer number $s \in \{1, \dots, n-1\}$, and $c_\delta \geq 0$, then*

$$\frac{1}{1 - \mathbf{c}^2} \leq 1 + \frac{\varepsilon_{r,s}^2}{\delta}, \quad (3.22)$$

for all $r \in \{0, \dots, s\}$, where

$$\varepsilon_{r,s} = \frac{\sum_{j=r}^s (j+1) a_j b_j \left(\max_M H_{j+1} \right)}{\sum_{j=r}^s (j+1) a_j b_j \left(\min_M H_j \right)}, \quad b_j = (j+1) \binom{n}{j+1},$$

a_j are nonnegative real numbers (with at least one nonzero), for all $j \in \{r, \dots, s\}$, and $\mathbf{c}$ is the center of gravity defined in (3.20).

Proof. From Proposition 1, we have

$$\mathbf{c}^2 = \frac{1}{\text{vol}(M)^2} \left(\int_M c_\delta dM \right)^2 \leq \frac{\varepsilon_{r,s}^2}{\text{vol}(M)^2} \left(\int_M s_\delta dM \right)^2. \quad (3.23)$$

On the other hand, the Cauchy–Schwarz inequality assures us that

$$\left(\int_M s_\delta dM \right)^2 \leq \left(\int_M s_\delta^2 dM \right) \text{vol}(M), \quad (3.24)$$

and, therefore, from (3.23) and (3.24), we get

$$\begin{aligned} (1 - \mathbf{c}^2) \left(1 + \frac{\varepsilon_{r,s}^2}{\delta} \right) &\geq 1 + \frac{\varepsilon_{r,s}^2}{\delta} - \frac{\varepsilon_{r,s}^2}{\text{vol}(M)} \left(\int_M s_\delta^2 dM \right) - \mathbf{c}^2 \frac{\varepsilon_{r,s}^2}{\delta} \\ &\geq 1 + \frac{\varepsilon_{r,s}^2}{\delta} - \frac{\varepsilon_{r,s}^2}{\text{vol}(M)} \left(\int_M s_\delta^2 dM \right) \\ &\quad - \left(\frac{1}{\text{vol}(M)} \int_M c_\delta^2 dM \right) \frac{\varepsilon_{r,s}^2}{\delta} \\ &= 1 + \varepsilon_{r,s}^2 \left(\frac{1}{\delta} - \frac{1}{\delta \text{vol}(M)} \int_M (\delta s_\delta^2 + c_\delta^2) dM \right) = 1. \end{aligned} \quad (3.25)$$

Hence, from (3.25), we obtain (3.22). $\square$

So, we are in position to prove our second result.

Theorem 2. *Let $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ be an orientable closed hypersurface immersed in a Riemannian space form $\overline{M}_\delta^{n+1}$ with constant sectional curvature $\delta > 0$, such that $x(M)$ is contained in a convex ball of radius $\frac{\pi}{4\sqrt{\delta}}$. If $H_{s+1} > 0$, for some integer number $s \in \{1, \dots, n-1\}$, then*

$$\lambda_1^{\mathcal{L}_{r,s}} \leq \frac{1}{\text{vol}(M)} (\delta + \varepsilon_{r,s}^2) \sum_{j=r}^s (j+1) a_j b_j \int_M H_j dM, \quad (3.26)$$

for all $r \in \{0, \dots, s\}$, where

$$\varepsilon_{r,s} = \frac{\sum_{j=r}^s (j+1) a_j b_j \left(\max_M H_{j+1} \right)}{\sum_{j=r}^s (j+1) a_j b_j \left(\min_M H_j \right)}, \quad b_j = (j+1) \binom{n}{j+1}.$$

Here, $\lambda_1^{\mathcal{L}_{r,s}}$ is the first positive eigenvalue of the operator $\mathcal{L}_{r,s}$ defined in (1.1) and a_j are nonnegative real numbers (with at least one nonzero) for all $j \in \{r, \dots, s\}$. Moreover, the equality occurs in (3.26) if, and only if, $x(M)$ is a sphere of $\overline{M}_\delta^{n+1}$ and x is its inclusion as a geodesic sphere.

Proof. As in the proof of Theorem 1, for all $1 \leq i \leq n+1$, let us consider $f_i = \frac{s_\delta(d)}{d} x_i$. From (3.7) and (3.21), considering the center of gravity $\mathbf{c}$ defined in (3.20), we have

$$\int_M \frac{c_\delta - \mathbf{c}}{\sqrt{\delta}} dM = 0 \quad \text{and} \quad \int_M f_i dM = 0, \quad i \in \{1, \dots, n+1\}. \quad (3.27)$$

Thus, we can use (3.9) jointly with Lemma 2 to obtain

$$\begin{aligned}
 & \lambda_1^{\mathcal{L}_{r,s}} \int_M \left(\frac{c_\delta - \mathbf{c}}{\sqrt{\delta}} \right)^2 dM \\
 & \leq \sum_{j=r}^s (j+1) a_j \int_M \left\langle P_j \left(\nabla \left(\frac{c_\delta - \mathbf{c}}{\sqrt{\delta}} \right) \right), \nabla \left(\frac{c_\delta - \mathbf{c}}{\sqrt{\delta}} \right) \right\rangle dM \\
 & = \frac{1}{\delta} \sum_{j=r}^s (j+1) a_j \int_M \langle P_j(\nabla(c_\delta - \mathbf{c})), \nabla(c_\delta - \mathbf{c}) \rangle dM \\
 & = \delta \sum_{j=r}^s (j+1) a_j \int_M \langle P_j(X^\top), X^\top \rangle dM, \tag{3.28}
 \end{aligned}$$

where in the last equality we used the fact that $\nabla(c_\delta - \mathbf{c}) = -\delta X^\top$.

Now, from (3.12), (3.13) and (3.28), we get

$$\begin{aligned}
 \lambda_1^{\mathcal{L}_{r,s}} \int_M \left(s_\delta^2 + \frac{(c_\delta - \mathbf{c})^2}{\delta} \right) dM & \leq \sum_{j=r}^s (j+1) a_j \int_M \sum_{i=1}^{n+1} \langle P_j(\nabla f_i), \nabla f_i \rangle dM \\
 & \quad + \delta \sum_{j=r}^s (j+1) a_j \int_M \langle P_j(X^\top), X^\top \rangle dM \\
 & = \sum_{j=r}^s (j+1) a_j b_j \int_M H_j dM. \tag{3.29}
 \end{aligned}$$

On one hand, from (3.20), we have

$$\int_M \left(s_\delta^2 + \frac{(c_\delta - \mathbf{c})^2}{\delta} \right) dM = \frac{1}{\delta} \text{vol}(M)(1 - \mathbf{c}^2), \tag{3.30}$$

and, therefore, from (3.29) and (3.30) we obtain

$$\lambda_1^{\mathcal{L}_{r,s}} \leq \frac{\delta}{\text{vol}(M)} \left(\frac{1}{1 - \mathbf{c}^2} \right) \sum_{j=r}^s (j+1) a_j b_j \int_M H_j dM. \tag{3.31}$$

On the other hand, Proposition 2 assures us that

$$\frac{\delta}{1 - \mathbf{c}^2} \leq \delta + \varepsilon_{r,s}^2. \tag{3.32}$$

Hence, from (3.31) and (3.32), we obtain (3.26).

Moreover, if the equality occurs in (3.26) then, occurs the equality in Proposition 1 which implies $-\langle X, N \rangle = |X||N|$. Therefore, d is constant on $x(M)$ and, hence, $x(M)$ is a sphere of $\mathbb{R}^{n+1}$ around p_0 and x is its inclusion as a geodesic sphere. $\square$

In our last result, considering a hypersurface M^n immersed in a hyperbolical space $\overline{M}_\delta^{n+1}$, we obtain a sharp upper bound for the first positive eigenvalue of the operator $\mathcal{L}_{r,s} - q : C^\infty(M) \rightarrow C^\infty(M)$, where $\mathcal{L}_{r,s}$ is defined in (1.1) and

$$q = - \sum_{j=r}^s (j+1) a_j [\operatorname{tr}(\delta P_j) + \operatorname{tr}(A^2 P_j)].$$

Theorem 3. Let $x : M^n \rightarrow \overline{M}_\delta^{n+1}$ be an orientable closed hypersurface immersed in a Riemannian space $\overline{M}_\delta^{n+1}$ with constant sectional curvature $\delta < 0$. If $H_{s+1} > 0$, for some integer number $s \in \{1, \dots, n-1\}$, then

$$\lambda_1^{\mathcal{L}_{r,s}-q} \leq \frac{\sum_{j=r}^s (j+1) a_j b_j \int_M (\varepsilon_{r,s}^2 \min_M H_j - H_1 H_{j+1}) s_\delta^2 dM}{\int_M s_\delta^2 dM}, \quad (3.33)$$

for all $r \in \{0, \dots, s\}$, where

$$\varepsilon_{r,s} = \frac{\sum_{j=r}^s (j+1) a_j b_j \left(\max_M H_{j+1} \right)}{\sum_{j=r}^s (j+1) a_j b_j \left(\min_M H_j \right)}, \quad b_j = (j+1) \binom{n}{j+1}.$$

Here, $\lambda_1^{\mathcal{L}_{r,s}-q}$ is the first positive eigenvalue of the operator $\mathcal{L}_{r,s}-q$ and a_j are nonnegative real numbers (with at least one nonzero) for all $j \in \{r, \dots, s\}$. Moreover, the equality occurs in (3.33) if, and only if, $x(M)$ is a sphere of $\overline{M}_\delta^{n+1}$ and x is its inclusion as a geodesic sphere.

Proof. Since $H_{s+1} > 0$, Lemma 3 guarantees that L_j is elliptic and $H_j > 0$ for all $j \in \{0, \dots, s\}$. Thus, $\mathcal{L}_{r,s}-q$ is an elliptic operator and its first positive eigenvalue $\lambda_1^{\mathcal{L}_{r,s}-q}$ admits a similar characterization that is given in (3.9).

As in the proof of Theorem 1, for all $i \in \{1, \dots, n+1\}$, we consider $f_i = \frac{s_\delta(d)}{d} x_i$ and from (3.7) we have that $\int_M f_i dM = 0$. Then, from (3.9), Lemma 2 and (2.1), we obtain

$$\begin{aligned} & \lambda_1^{\mathcal{L}_{r,s}-q} \int_M f_i^2 dM \\ & \leq - \int_M \left(\sum_{j=r}^s (j+1) a_j f_i L_j(f_i) - q f_i^2 \right) dM \\ & = \sum_{j=r}^s (j+1) a_j \int_M \langle P_j(\nabla f_i), \nabla f_i \rangle dM \\ & \quad + \sum_{j=r}^s (j+1) a_j \int_M \left(b_{j+1} H_{j+2} - \frac{n b_j}{j+1} H_1 H_{j+1} - \delta b_j H_j \right) f_i^2 dM. \end{aligned} \quad (3.34)$$

Summing on i from 1 to $n+1$ in (3.34) and using (3.12), we get

$$\begin{aligned} & \lambda_1^{\mathcal{L}_{r,s}-q} \int_M s_\delta^2 dM \\ & \leq \sum_{j=r}^s (j+1) a_j \int_M \sum_{i=1}^{n+1} \langle P_j(\nabla f_i), \nabla f_i \rangle dM \\ & \quad + \sum_{j=r}^s (j+1) a_j \int_M \left(b_{j+1} H_{j+2} - \frac{n b_j}{j+1} H_1 H_{j+1} - \delta b_j H_j \right) s_\delta^2 dM. \end{aligned} \quad (3.35)$$

On the other hand, from (3.13) and (3.16) we see that the first term of the right side of (3.35) can be estimated as:

$$\begin{aligned}
 & \sum_{j=r}^s (j+1)a_j \int_M \sum_{i=1}^{n+1} \langle P_j(\nabla f_i), \nabla f_i \rangle dM \\
 & \leq \delta \sum_{j=r}^s (j+1)a_j b_j \int_M s_\delta^2 H_j dM \\
 & \quad + \sum_{j=r}^s (j+1)a_j b_j \left(\max_M H_{j+1} \right) \varepsilon_{r,s} \int_M s_\delta^2 dM. \tag{3.36}
 \end{aligned}$$

But, considering (3.36) into (3.35), we conclude

$$\begin{aligned}
 \lambda_1^{\mathcal{L}_{r,s}-q} \int_M s_\delta^2 dM & \leq \sum_{j=r}^s (j+1)a_j b_j \left(\min_M H_j \right) \varepsilon_{r,s}^2 \int_M s_\delta^2 dM \\
 & \quad + \sum_{j=r}^s (j+1)a_j \int_M \left(b_{j+1} H_{j+2} - \frac{nb_j}{j+1} H_1 H_{j+1} \right) s_\delta^2 dM. \tag{3.37}
 \end{aligned}$$

So, using the fact that $H_{j+2} \leq H_1 H_{j+1}$, with equality at umbilical points (cf. [3, p. 392]), from (3.37), we have

$$\begin{aligned}
 \lambda_1^{\mathcal{L}_{r,s}-q} \int_M s_\delta^2 dM & \leq \sum_{j=r}^s (j+1)a_j b_j \left(\min_M H_j \right) \varepsilon_{r,s}^2 \int_M s_\delta^2 dM \\
 & \quad + \sum_{j=r}^s (j+1)a_j \int_M \left(b_{j+1} - \frac{nb_j}{j+1} \right) H_1 H_{j+1} s_\delta^2 dM \\
 & = \sum_{j=r}^s (j+1)a_j b_j \int_M \left(\varepsilon_{r,s}^2 \min_M H_j - H_1 H_{j+1} \right) s_\delta^2 dM, \tag{3.38}
 \end{aligned}$$

where in last equality, it was used that $b_{j+1} - \frac{nb_j}{j+1} = -b_j$. Thus, passing to the quotient (3.38) by $\int_M s_\delta^2 dM$, we obtain (3.33).

If equality occurs in (3.33), we can proceed as in the proofs of the previous theorems to conclude that $x(M)$ must be a sphere of $\overline{M}_\delta^{n+1}$ and x is its inclusion as a geodesic sphere. $\square$

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